ASSEMBLY INSTRUCTIONS

MICRODOCK

THE SELF-CHARGING STATION

A 3D-printed dock for Microduck. It finds the tag, sits down, scoots back under the hood and charges.

5 PRINTED PARTS	**~5½ h** PRINT, PLA
~10 min TO BUILD	**0** SCREWS, GLUE, TOOLS

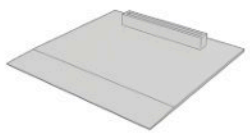


SCREWS ★ NO GLUE ★ NO TOOLS ★ ONE CABLE ★ MAGNETIC USB-C ★ FLOATING HOOD ★ PROTOTYPE ★ NO SCREWS ★ NO GLUE ★ NO

MicroDock v2 · for Microduck · rendered from the CAD and the simulation

WHAT YOU NEED

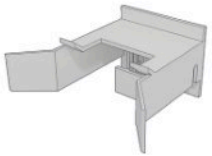
PRINT



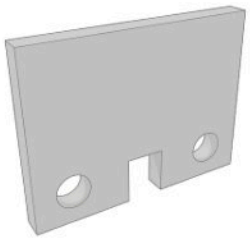
1X Base



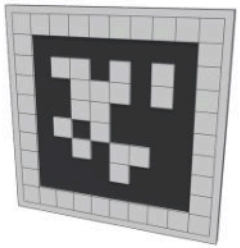
1X Board



1X Hood



1X Keeper



1X Tag



2X Filament pin, 12 mm



Plate 1 base

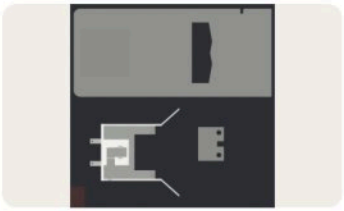


Plate 2 board, hood, keeper



Plate 3 tag (two colours)

PLA, white, plus black for the tag. The Bambu Lab P1S project sets everything: three plates, about 5 h 30 min and 240 g. The hood prints upside down on supports, already set in the project. The two pins are just pieces of 1.75 mm filament: snap them off your spool.

GET



1X magnetic USB-C adapter (a pair): the adapter goes in the hood, the tip in Microduck

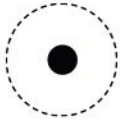


1X right-angle USB-C cable: the hood is shaped for this head (UGREEN 100 W)



1X USB-C charger: any phone charger will do

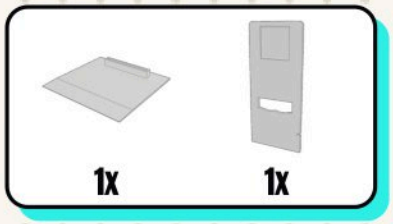
CABLE CHECK 1:1

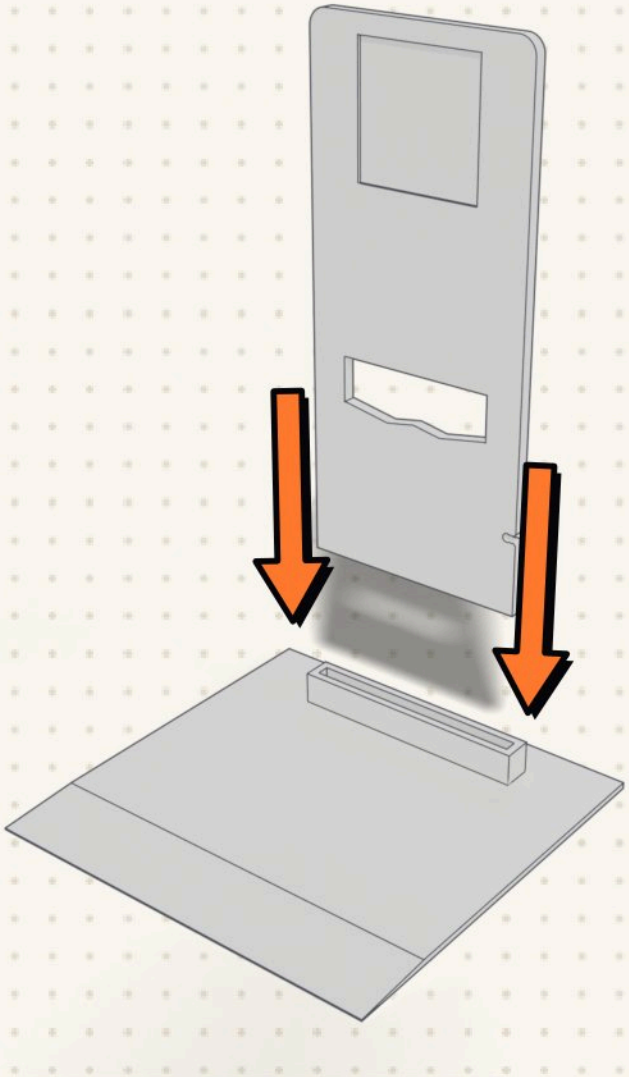


Print this page at 100 %. The cable should just cover the black disc (4 mm). The board's clip is made for it.

NO TOOLS NEEDED!

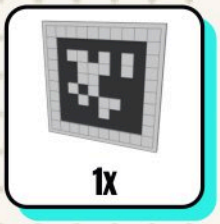
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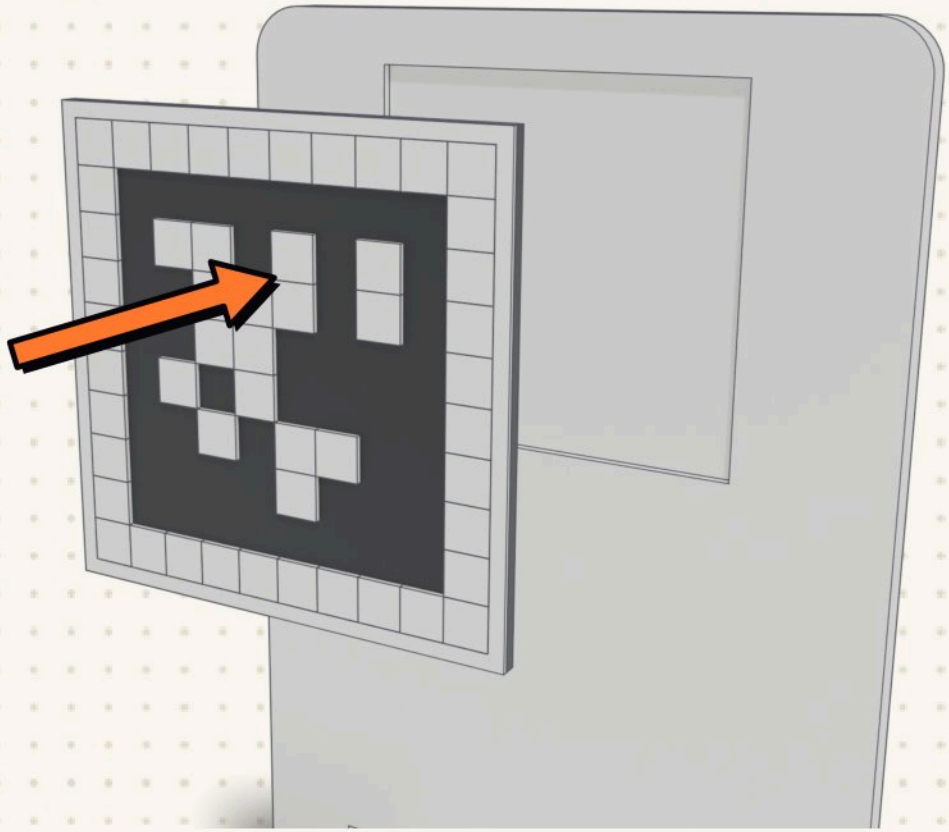




Push the board down into the base's socket, the tag pocket facing out.

2

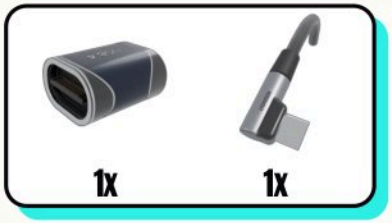




Press the tag into its pocket on the board, white cells out.

BUILD FIRST: HOOD + CABLE

3



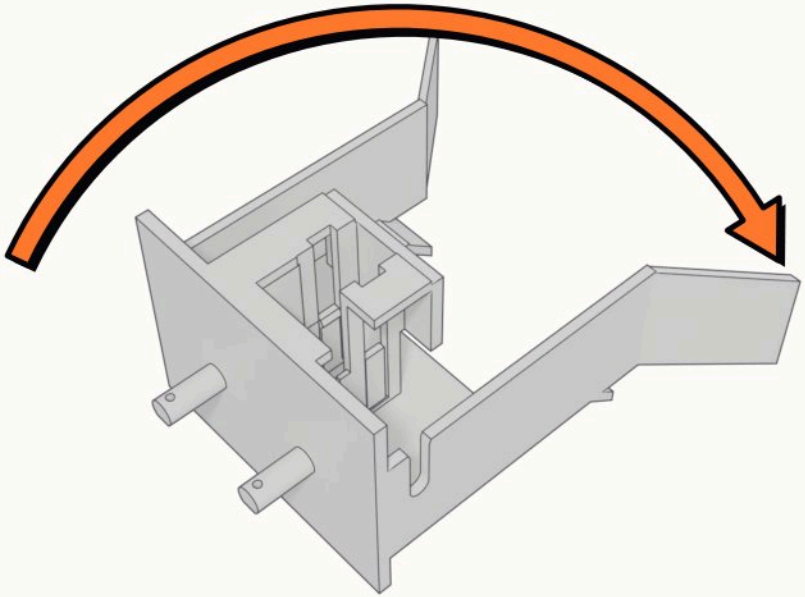


Plug the right-angle cable into the magnetic adapter.

4





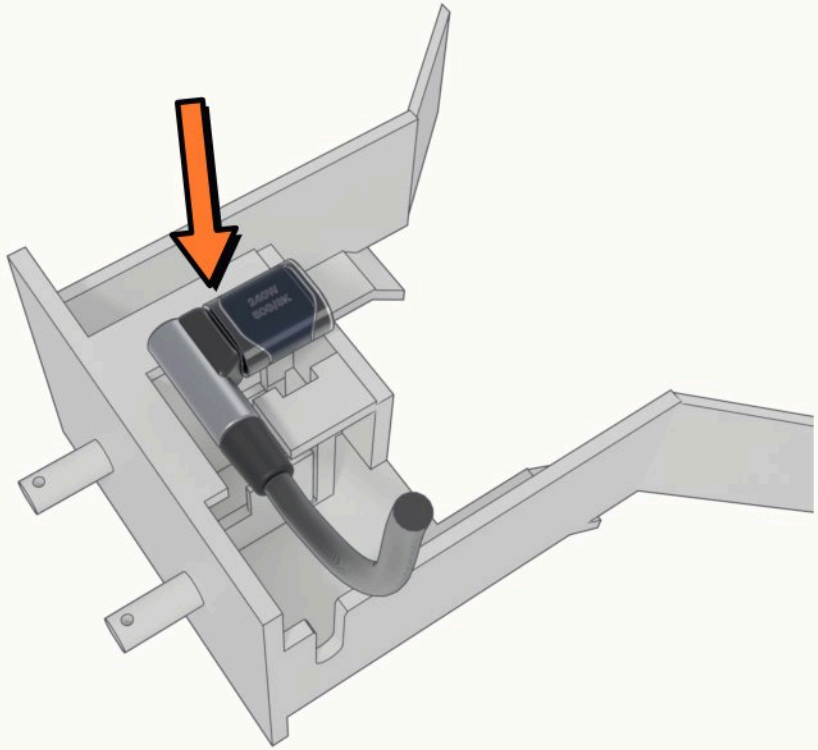


Turn the hood upside down.

BUILD FIRST: HOOD + CABLE (CONTINUED)

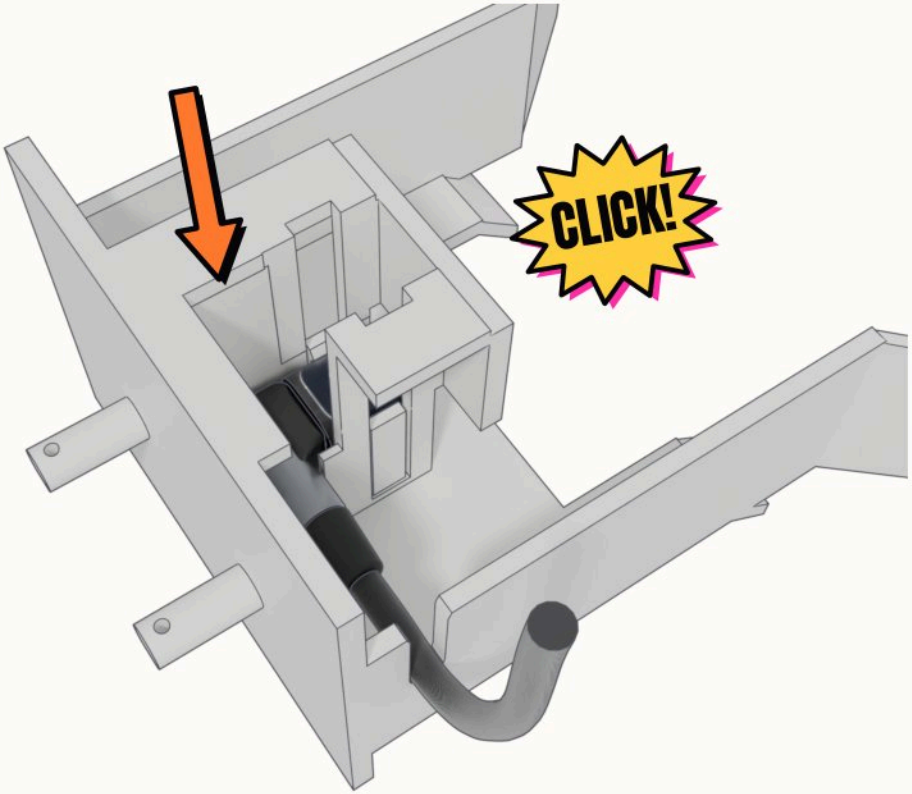
5





Lower the pair into the hood: the adapter into its pocket, magnetic face toward the hood's flared end, and the cable out through the slot in the side wall.

6

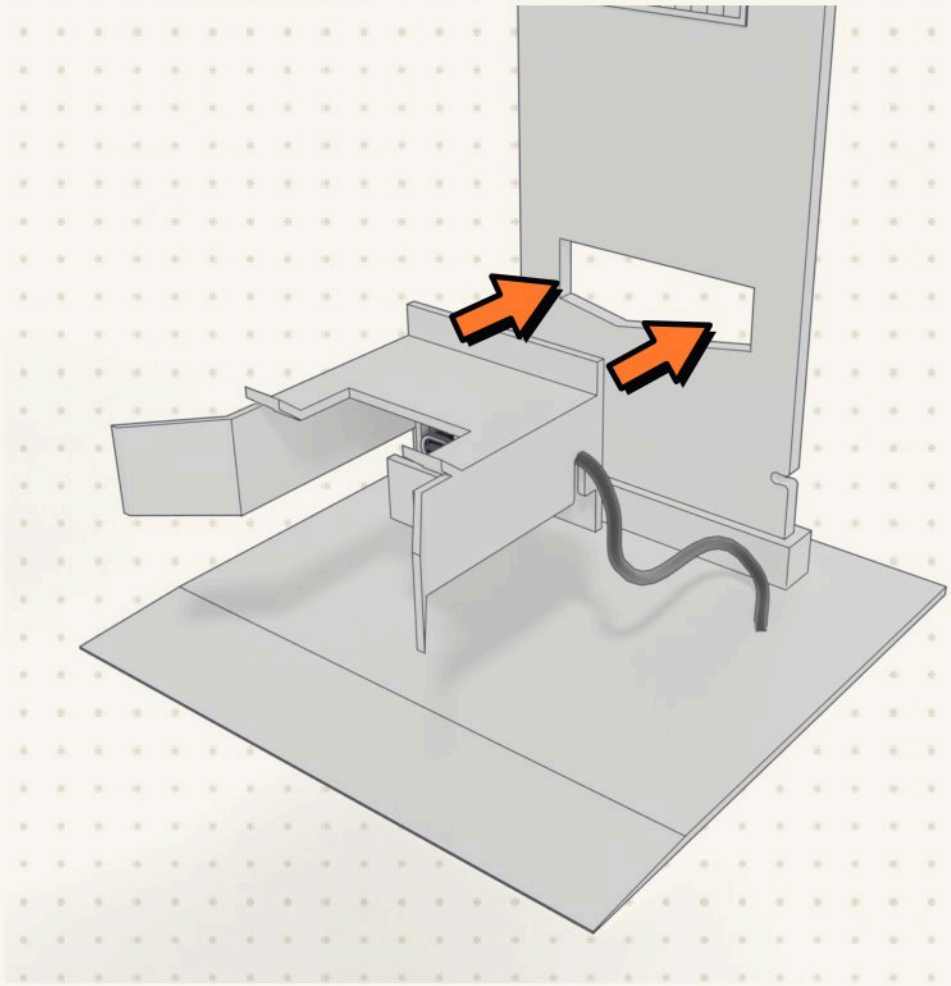


Press it down until it clicks.

> Two small clips in the pocket hold the adapter. To take it out again, pull the cable head out firmly.

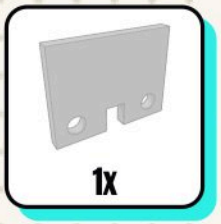
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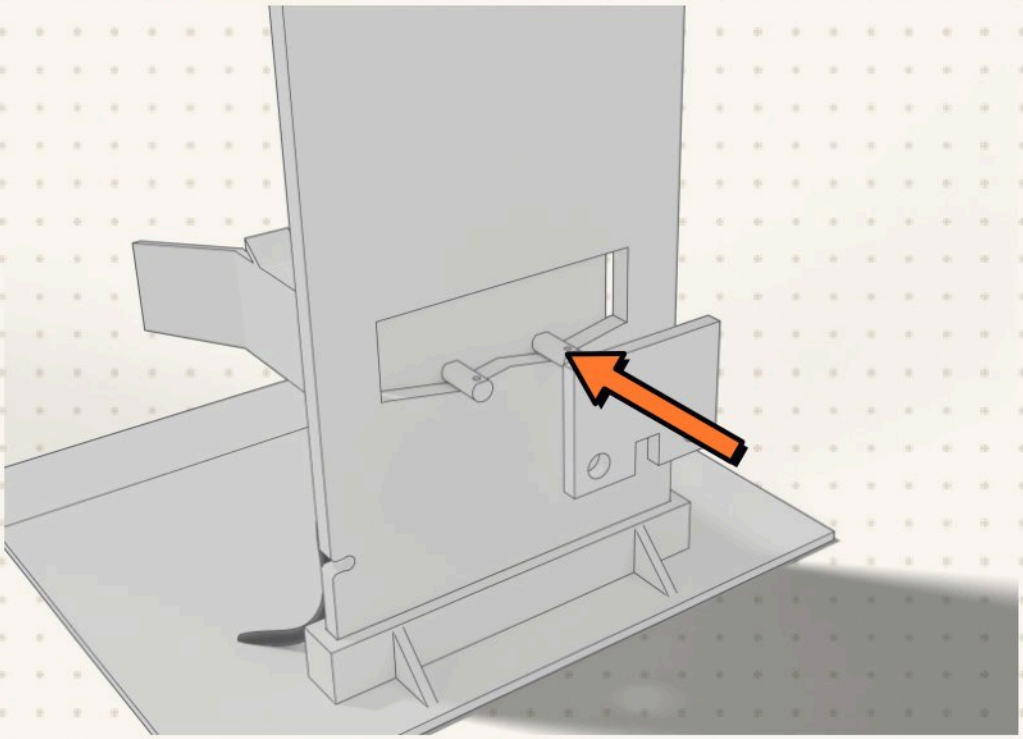




Turn the hood back over and push its two pegs through the window in the board, from the front.

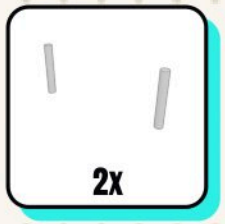
8

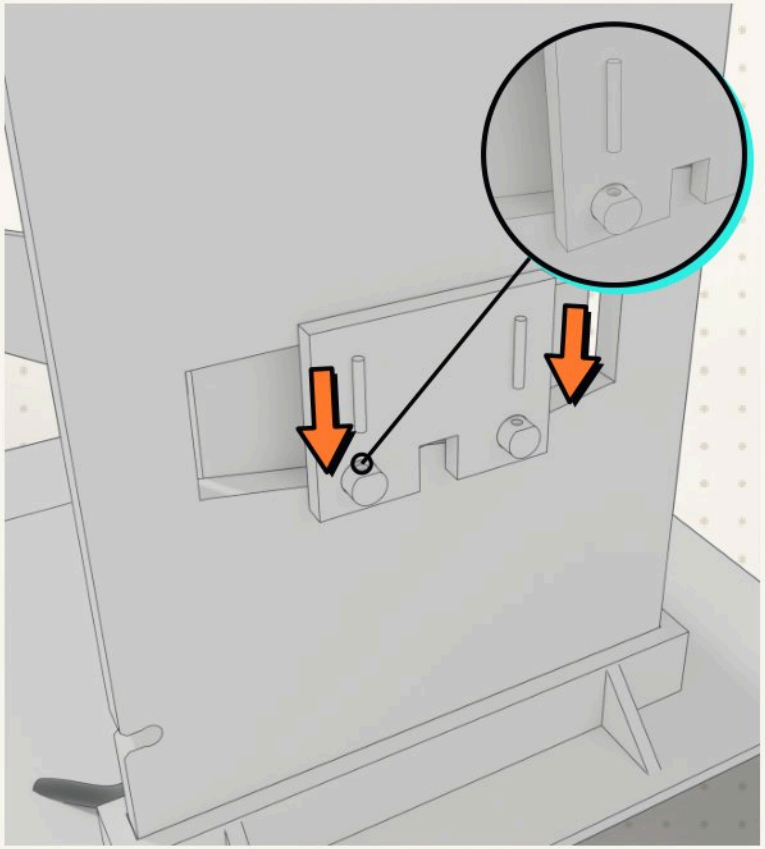




From behind, slide the keeper onto the two pegs.

9

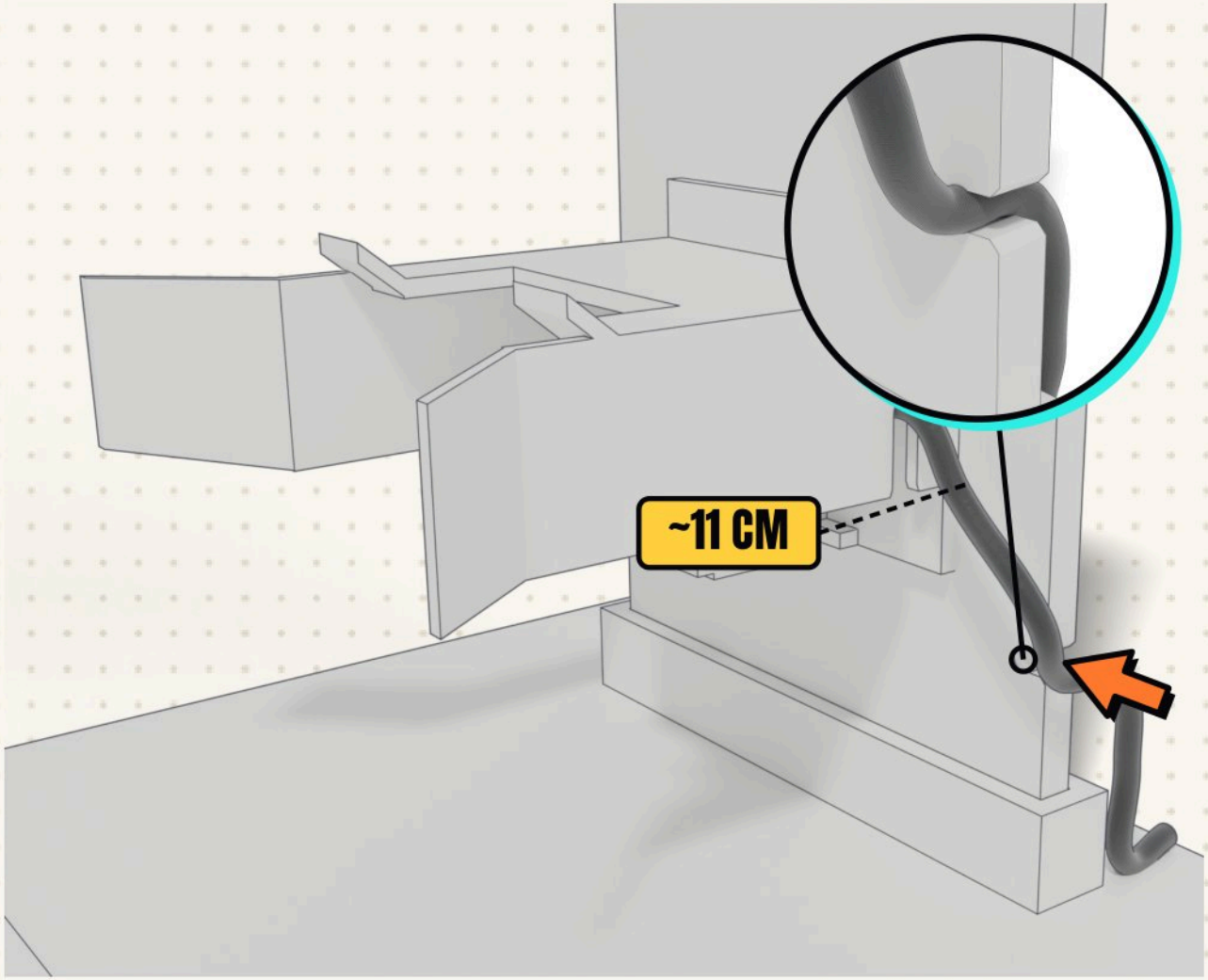




Lock it: push a pin down through the hole in each peg, right behind the keeper.

The pins keep the keeper on, and the keeper keeps the hood on the board.

10

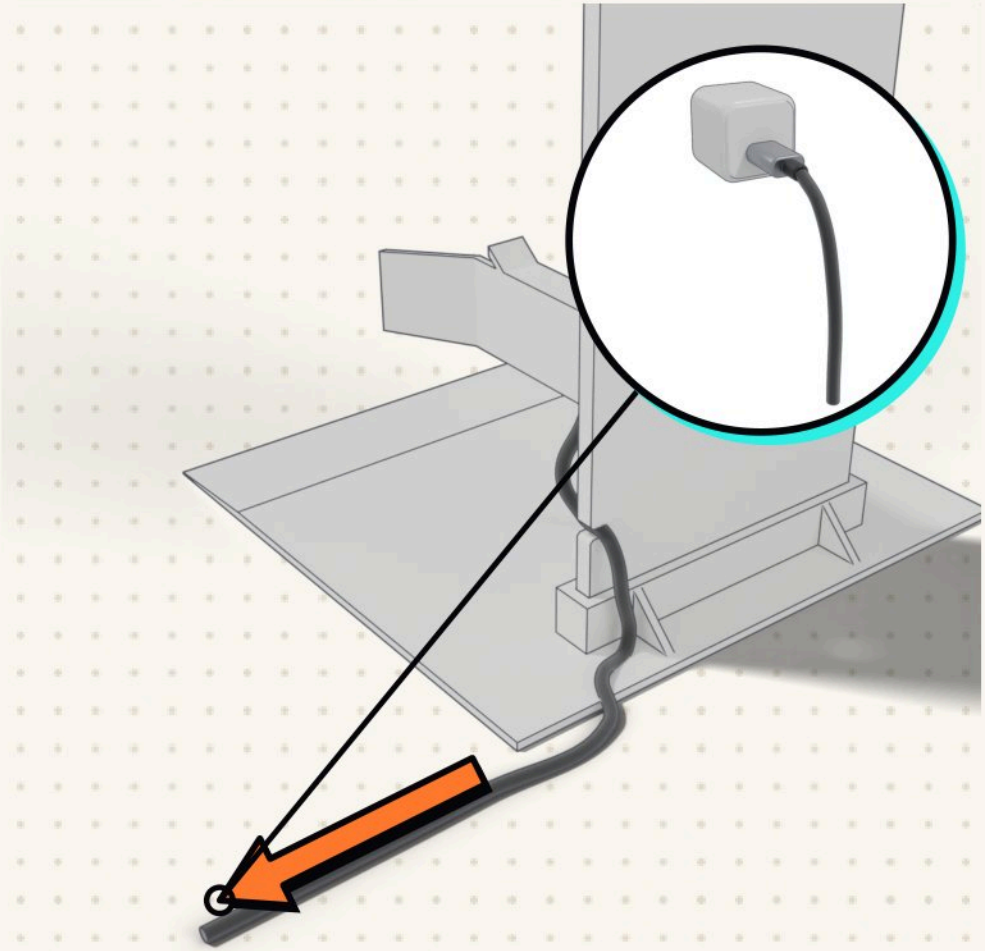


Press the cable into the clip low on the board's edge. Leave about 11 cm of cable between the hood and the clip.

The loop lets the hood float, and the clip keeps the cable's weight and any tug off the hood.

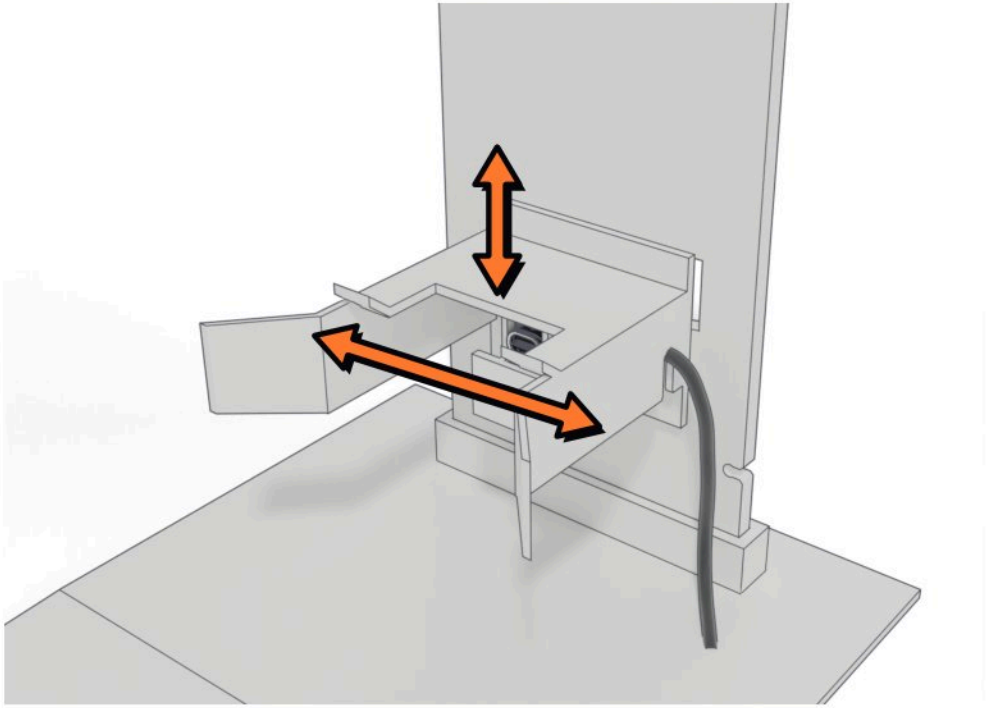
11





Run the cable down behind the board and plug it into a USB-C charger.

CHECK



The hood floats: lift it a little and slide it sideways. It must move freely, cable and all, and drop back down onto its seats. That's how it finds Microduck's back.

12



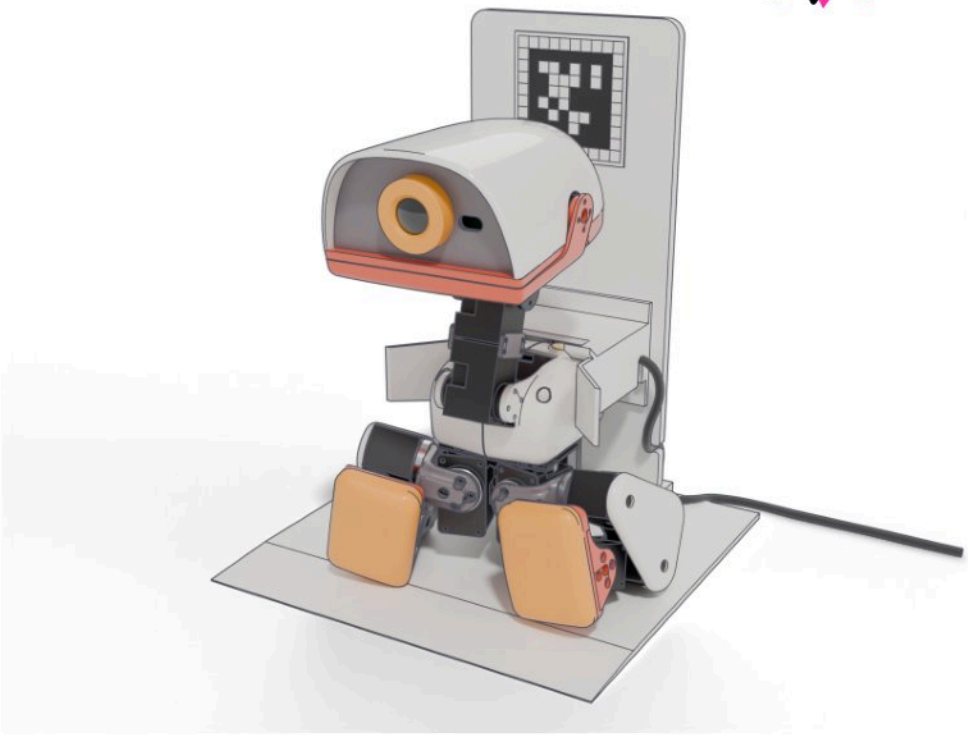
1x



On Microduck: push the magnetic tip into the USB-C port on its back. It stays there.

DONE!

CHARGE!



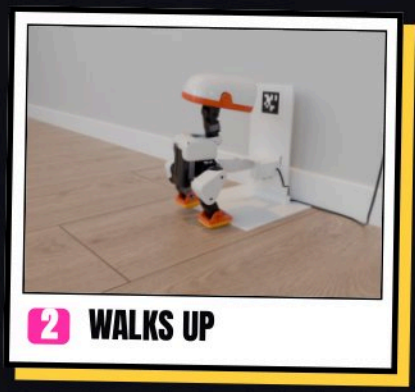
Stand the dock against a wall and stick the base to the floor with removable mounting tape: Microduck pushes the dock when it backs in and pulls it when it leaves.

Microduck spots the tag, walks up, sits down and scoots back under the hood. The magnets click, USB power comes on, and it dozes off: every servo off while it charges.

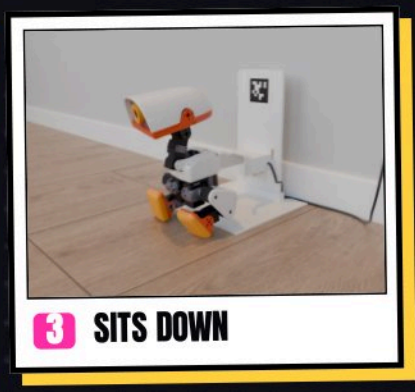
HOW IT DOCKS



1 SPOTS THE TAG



2 WALKS UP



3 SITS DOWN



4 SCOOTS BACK



5 CLICK: CHARGING

GOOD TO KNOW

- **Check your Microduck first.** Recent units have the USB-C port on the back, just above the battery. The hood is shaped for that.
- **Taking the adapter out:** pull the cable head down firmly. The fingers let go.
- **Keep the dock in place:** against a wall, the base taped down. Docking pushes it back by a few newtons and leaving pulls it forward; on its own it would slide on a smooth floor.
- **The hood must move freely.** If it sticks, check that the keeper isn't pressed hard against the board. It doesn't re-centre by itself: leave it roughly in the middle.
- **Taking the hood off:** pull the two pins out and slide the keeper off.
- **Other cables:** the hood fits the UGREEN head and the clip a 4 mm cable. Both are parameters in `source/`.
- **Prototype.** The docking moves run in simulation so far. Microduck's own software doesn't have them yet.

HAPPY DOCKING!

NT ★ PIN ★ CLIP ★ PLUG ★ DOCK ★ DOZE ★ PRINT ★ PIN ★ CLIP ★ PLUG ★ DOCK ★ DOZE ★